Tutorial 13 Thermodynamics Systems Suggested Solutions

D1 **(a)** **(i)**

$$pV = nRT \quad \rightarrow n = \frac{pV}{RT}$$

Since volume and temperature are fixed, $\Delta n = \dfrac{(\Delta p)V}{RT}$

$$\Delta n = \frac{(\Delta p)V}{RT} = \frac{(3.23 - 2.62)\times 10^5 (0.0120)}{(8.31)(25 + 273.15)} = 0.29544 = 0.295 \text{ mol}$$

(ii) To supply all 4 tyres, total amt of gas needed $= 4 \times 0.295 = 1.18$ mol

The accompanying change in pressure

$$\Delta p = \frac{(\Delta n)RT}{V} = \frac{(1.18)(8.31)(25 + 273.15)}{(0.0108)} = 2.7070 \times 10^5 \text{ Pa}$$

This will result in a new pressure of

$= (8.72 - 2.7070)\times 10^5$

$= 6.01\times 10^5$ Pa $> 3.23 \times 10^5$ Pa (shown)

Hence, there is more than enough air to supply four tyres.

(b) **(i)** Internal energy of an ideal gas is purely kinetic.

$$U = \frac{3}{2}NkT = \frac{3}{2}(1)\left(1.38\times 10^{-23}\right)(25 + 273.15) = 6.17\times 10^{-21} \text{ J (shown)}$$

(ii) Internal energy of one mole

$$U = N \times U = \left(6.02\times 10^{23}\right)\left(6.17\times 10^{-21}\right) = 3710 \text{ J}$$

(iii) Increase in internal energy of the air

$$\Delta U = \Delta n \times (U \text{ of one mol}) = (0.295)(3710) = 1090 \text{ J}$$

D2 **(a)**

- Both water and vapour molecules have freedom of motion and move about in continuous random motion.
- Attractive intermolecular forces exist between water molecules. Hence they are more closely spaced and their freedom of movement is restricted by the volume of water itself.
- Weak (or negligible) attractive intermolecular forces exist between water vapour molecules. Hence, their movement is less restricted.
- At the same temperature, both water molecules and water vapour molecules have the same root mean square speed.

(b)

- The internal energy of a system is the sum of the microscopic kinetic energy, due to the random motion of the molecules, and the microscopic potential energy, associated with the intermolecular forces of the system.

(c)

- Since both water and water vapour are at the same temperature, their molecules have the same mean microscopic kinetic energy. Hence, their kinetic energy per unit mass are the same.
- The water vapour molecules are spaced very far apart and have much fewer intermolecular interactions as compared to liquid water molecules. Hence, the potential energy of the water vapour molecules is larger (close to zero) than that of the liquid water molecules, which are spaced close together and hence have a lower potential energy (more negative).
- Hence, the internal energy per unit mass of liquid water is lower than that of water vapour.

Note: The potential energy between molecules is similar to that of gravitational potential energy. If a gas is taken to have zero microscopic potential energy, then the corresponding liquid and solid must have lower/negative microscopic potential energy because work must be done or heat must be supplied to separate the molecules.

(d)

- During a phase change, there is no change in temperature, and no change in the average kinetic energy of the molecules.
- However, there is a change in the internal energy of a substance because of the change in potential energy of the molecules. During melting, the intermolecular bonds are weakened. During vaporisation, the intermolecular bonds are broken.
- As such, the change in microscopic potential energy (and hence internal energy) of the molecules during the vaporisation process is greater than that during the melting process.
- In addition, during vaporisation, work must be done when the gas expands against atmospheric pressure.
- Thus, the specific latent heat of vaporisation is greater than that of fusion

D3 **(a)** **(i)** The two bodies at thermal equilibrium have the <u>same temperature</u>.

(ii) There is no <u>net</u> flow (or transfer) of thermal energy from one body to another.

Comment: if the word "net" is missing, candidates do not get full credit.

(b) **(i)** No.
The ice and the surrounding are at different temperature. Hence there is thermal energy being transferred from the surroundings to the ice.

(b) **(ii)** 5.0 minutes = 300 s.

electrical energy supplied $+$ energy supplied by surrounding

$=$ energy to melt ice

$$IVt + R\,t = \Delta mL$$

$$(6.3)(12.0)(300) + 300R = (114.0 - 32.4)(330)$$

$$R = 14.16 = 14.2 \text{ W} \text{ (2 or 3 s.f.)}$$

D4 **(a)** $pV = nRT$

Considering point A, $n = \dfrac{pV}{RT} = \dfrac{(4\times10^5)(0.002)}{(8.31)(600)} = 0.16$ mol

(b) $\frac{pV}{T} = nR = (0.16)(8.31) = 1.3$

If the gas behaves as an ideal gas, all the graphs should satisfy the above relationship that $\frac{pV}{T} = 1.3$

Consider 2 data points on each graph:

For $T = 300$ K, $\frac{(4\times10^5)(0.001)}{300} = 1.3$

$\frac{(2\times10^5)(0.002)}{300} = 1.3$

For $T = 600$ K, $\frac{(4\times10^5)(0.002)}{600} = 1.3$

$\frac{(2\times10^5)(0.004)}{600} = 1.3$

For $T = 900$ K, $\frac{(6\times10^5)(0.002)}{900} = 1.3$

$\frac{(3\times10^5)(0.004)}{900} = 1.3$

For $T = 1200$ K, $\frac{(8\times10^5)(0.002)}{1200} = 1.3$

$\frac{(4\times10^5)(0.004)}{1200} = 1.3$

(c) **(i)** $Q = C\Delta T = (3.33)(1200 - 600) = 2000$ J

(ii) $Q = C\Delta T = (4.66)(1200 - 600) = 2800$ J

(d) $U \propto T$

Since $T_B = 2T_A$, $U_B = 2U_A = 4000$ J

Point C is at the same temperature as point B. Since internal energy is directly proportional to temperature, the internal energy at C is also 4000 J.

(e) Work done = Area under *p*-*V* graph = $(4 \times 10^5)(0.004 - 0.002) = 800$ J

(f) Point D is at the same temperature as point A. Since internal energy is directly proportional to temperature, the internal energy at D is also 2000 J.

Note:

If we consider the gas as monatomic atom and apply $U = \frac{3}{2}PV$ to solve (c) and (d), the answers obtained will not be consistent with the date provided.

The examiners actually considered the gas as a diatomic molecule and applied $U = \frac{5}{2}PV$.

Hence, students are advised to use the first law of thermodynamic to solve questions and use $U = \frac{3}{2}PV$ only when the first law of thermodynamic cannot be applied.

D5 **(a)** $\Delta U = Q + W$

$\Delta U = 0 + (-600)$

(b) -600 J

(c) The temperature of the gas decreases.

D6 **(a)** $Q = mc\Delta T = (5.6)(0.73)(1) = 4.09$ J

(b) **(i)** Since pressure is constant,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$V_2 = \frac{(4.6 \times 10^3)}{280}(281) = 4616 \text{ cm}^3$$

Change in volume of the gas = 4616 – 4600 = 16 cm^3

(ii) Work done by the gas $= p\Delta V$

$$= (1.0 \times 10^5)(16 \times 10^{-6})$$

$$= 1.6 \text{ J}$$

(c) **(i)** The First Law of Thermodynamics states that the increase in the internal energy of a system is equal to the sum of the heat supplied to the system and the work done on the system, and the internal energy of a system depends only on its state.

(ii) For the process occurring at constant volume in **(a)**, $W = 0$

$\Delta U = Q = 4.09$ J

For the process occurring at constant pressure in **(b),** the increase in internal energy is the same as that in **(a)** since $\Delta U \propto \Delta T$ and $\Delta T = 1$ K for both cases.

Hence, for process **(b),**

$$\Delta U = Q_b + W$$

$$4.09 = Q_b + (-1.6)$$

$$Q_b = 5.69 \text{ J}$$

Note:

$$n = \frac{5.6}{28} = 0.20 \text{ mol}$$

For constant volume, $Q = \Delta U = \frac{5}{2}nR\Delta T = \frac{5}{2} \times 0.20 \times 8.31 \times 1 = 4.16$ J

By considering nitrogen gas as a diatomic molecule, N_2, we obtained an answer that is slightly different from that in **(a)**.

Considering nitrogen gas as a monatomic atom results in factor of $\frac{3}{2}$ instead of $\frac{5}{2}$

D7 **(a)** $pV = nRT$

$$n = \frac{(2.4\times10^5)(5.0\times10^{-4})}{(8.31)(290)} = 0.050 \text{ mol}$$

(b) **(i)** Magnitude of work done $= p|\Delta V|$

$$= (2.4\times10^5)\left[(14.4-5.0)\times10^{-4}\right]$$

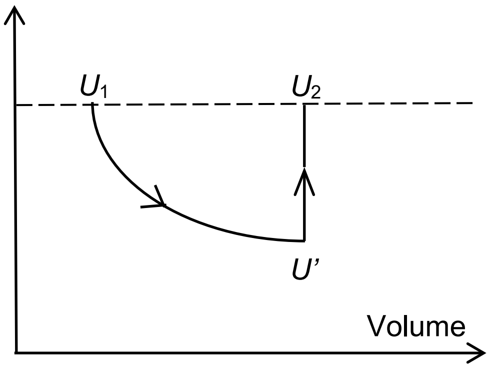
$$= 226 \text{ J}$$

(ii) Change in internal energy during one complete cycle = 0 J

(c)

Change	Work done on gas / J	Heat supplied to gas / J	Increase in internal energy / J
X → Y	**- 230**	+ 570	**+ 340**
Y → Z	+ 540	0	**+ 540**
Z → X	0	**- 880**	**- 880**

D8 **(a)**



$U_1 \rightarrow U'$, work done by gas is W. Work done on gas is $-W$.

Section of cycle	Q / J	W / J	ΔU/J
$U_1 \rightarrow U'$	0	- W	- W
$U' \rightarrow U_2$	Q	0	Q

(b)

Section of cycle	Q /J	W / J	ΔU/J
AB	280	**0**	**280**
BC	20	170	**190**
CD	–400	**0**	**–400**
DA	**0**	–70	**–70**

D9 Since not all the ice melted, the equilibrium temperature is 0°C.

Heat lost by water $=$ Heat gained by ice $+$ Heat required to melt $(M-m)$ of ice

$$m_{\text{water}}c_{\text{water}}\Delta T = Mc_{\text{ice}}\Delta T + (M-m)\,l_f$$

$$(1.000)(4190)(20) = M(2050)(20) + (M-m)(333000)$$

$$m = 1.123M - 0.252$$

D10 **(a)** Since power is constant,

$$P = mc\frac{\Delta T}{\Delta t} = m(1800)\left(\frac{80-20}{200-0}\right) = 540m$$

X takes 200 s to melt.

$$Pt = ml_f$$

$$(540m)(200) = ml_f$$

$$l_f = 108000$$

$$= 1.08\times10^{5}\ \text{J kg}^{-1}$$

(b) The temperature of X increases over 100 s.

$$Pt_2 = mc_l(\Delta T)_2$$

$$(540m)(100) = mc_l(110-80)$$

$$c_l = 1800\ \text{J kg}^{-1}\ \text{K}^{-1}$$

D11 **(a)** This is to allow for heat loss to be eliminated.

Since the rate of heat loss is dependent on the temperature difference between the system and the surroundings (which is assumed to be constant), using two different heating rates would give the same amount of heat loss for the same amount of time. Rate of heat loss can therefore be eliminated from the equation.

(b)

$$P = \frac{m}{t}l_v + \frac{H}{t}$$

$$(140) = \frac{(14.1)}{(5\times60)}l_v + \frac{H}{t} \quad \rightarrow (1)$$

$$(95) = \frac{(8.2)}{(5\times60)}l_v + \frac{H'}{t} \quad \rightarrow (2)$$

Since $\dfrac{H}{t} = \dfrac{H'}{t}$

Solving for simultaneous equations

$l_v = 2300\ \text{J g}^{-1} = 2.3\times10^{6}\ \text{J kg}^{-1}$

C1 **(a)** Heat of combustion = 4.03×10^{7} J L^{-1}

Volume of fuel needed in 1 hour = $\dfrac{7.89\times10^{3}\times2500\times60}{4.03\times10^{7}}$ = 29.4 L

(b) Mechanical work done per cycle = $Q_h - Q_c = 7.89 \times 10^3 - 4.58 \times 10^3 = 3.31 \times 10^3$ J

Power output = $\frac{W}{t} = \frac{3.31 \times 10^3}{60} \times 2500 = 1.38 \times 10^5$ W

(c) Power transferred out of the engine = $\frac{Q_c}{t} = \frac{4.58 \times 10^3}{60} \times 2500 = 1.91 \times 10^5$ W

C2 Cyclic process means $\Delta U = 0$

$\Delta U = Q + W$

$0 = Q_{net} + W_{net}$

$W_{net} = -Q_{net}$

Net Work done by gas = $Q_{net} = (Q_h - Q_c)$

$$\text{Efficiency } e = \frac{W}{Q_h} = \frac{Q_h - Q_c}{Q_h}$$

$$= 1 - \frac{Q_c}{Q_h}$$

$$= 1 - \frac{C_v(T_D - T_A)}{C_p(T_c - T_B)}$$

$$= 1 - \frac{1}{\gamma}\left(\frac{T_D - T_A}{T_C - T_B}\right)$$